MILITARY SPECIFICATION

SIGNAL, UNDERWATER SOUND, MARK 61 MOD 0 and MARK 64 MOD 0

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the requirements for the manufacture and acceptance of both the Signal, Underwater Sound (SUS), Mark 61 Mod 0 and Mark 64 Mod 0, referred to herein as the signal(s).

1.2 Classification. - The signals specified herein are of identical construction and vary only in explosive loading and are identified by different Mark and Mod.

1.2.1 Signal, Underwater Sound, Mk 61 Mod 0. - This signal contains a nominal explosive load of 1.1 ounces of tetryl and 1.8 pounds of TNT.

1.2.2 Signal, Underwater Sound, Mk 64 Mod 0. - This signal contains a nominal explosive load of 1.1 ounces of tetryl.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. - The following documents of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this document to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

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FSC 1361

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DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

DL 1332255	Pack, Shipping and Storage (For Six or Eight Signals, Underwater Sound)
DL 2116941	Explosive Section, Underwater Sound Signal, Mk 4 Mod 0, Loaded
DL 2116973	Explosive Section, Underwater Sound Signal, Mk 8 Mod 0, Loaded
DL 2127915	Pack, Shipping and Storage (For Six Signals, Underwater Sound)
DL 2302101	Arming and Firing Mechanism, Mk 33 Mod 0, Loaded
DL 2302108	Signal, Underwater Sound, Mk 61 Mod 0, Assembly, Loaded
DL 2302109	Signal, Underwater Sound, Mk 64 Mod 0, Assembly, Loaded
696819	Arming Wire Assembly

OTHER PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND
(Code Ident 53711)

WR 54/131	Weapons Requirement, Palletizing, Fleet Issue Unit Load, Signals, Underwater Sound
WR 54/167	Weapons Requirement, Palletizing, Fleet Issue Unit Load, Signals, Underwater Sound

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Item identification. - The signals specified herein are hydrostatically operated explosive loaded ordnance items designed to detonate at nominal water depths of 60 or 800 feet, selectable prior to launch, for the purpose of providing a high energy underwater sound source. The SUS Mk 61 Mod 0, DL 2302108, is composed of an arming and firing mechanism (A&F), Mk 33 Mod 0, DL 2302101; an explosive section, Mk 4 Mod 0, DL 2116941; and an arming wire assembly, Dwg. 696819. The SUS Mk 64 Mod 0, DL 2302109, is composed of an A&F, Mk 33 Mod 0, DL 2302101; an explosive section, Mk 8 Mod 0, DL 2116973; and an arming wire assembly, Dwg. 696819.

3.2 Government furnished property list. - The shipping containers and explosive components may be Government furnished when so specified in the contract or order.

3.3 Performance.

3.3.1 Time from water entry to detonation. - The signal, when launched from an aircraft, shall enter the water, sink to the preset depth and detonate. When detonation occurs, the time from water entry to detonation shall be as follows (see 4.5.1).

(a) Shallow depth setting: 4 ± 2 seconds (C1).

(b) Deep depth setting: 46 ± 7 seconds (C2).

3.3.2 Detonation quality. - The signal shall detonate high order as determined by the first bubble periods of Table I (see 4.5.2).

TABLE I. FIRST BUBBLE PERIODS

Depth setting	Mk 61		Mk 64	
	Period (msec)	Character-istic	Period (msec)	Character-istic
Shallow	121 ± 13	M101	42 ± 4	M103
Deep	19.5 ± 1.8	M102	6.8 ± .6	M104

3.4 Design and construction. - The signal shall be fabricated and assembled in accordance with the drawings, parts lists and other documents listed on DL 2302108 or DL 2302109, as specified in the contract or order.

3.5 First article. - When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.3).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. - The inspection requirements specified herein are classified as follows.

(a) First article inspection (see 4.3).

(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. - Unless otherwise specified in the contract or order, a first article sample of 80 signals shall be submitted for evaluation. The sample will be subjected by the Government to any or all of the examinations and tests specified in 4.5.1 and 4.5.2 of this specification and any or all requirements of the applicable drawings. The first article sample shall be manufactured using the same methods and materials prepared for production. Any production prior to acceptance of the first article sample shall be at the risk of the supplier.

4.3.1 First article performance test sampling. - Of the sample submitted, 20 each shall be tested under each Condition of Table II. Two defective units in any one of the samples of 20 shall reject the total first article sample for MAJOR characteristics. One CRITICAL defect shall reject the first article sample.

4.4 Quality conformance inspection. - The quality conformance inspection shall verify that all characteristics of Section 3, Requirements, and Section 5, Packaging, have been met.

4.4.1 Inspection lot. - The inspection lot shall be as defined in MIL-STD-105, shall not exceed 3200 signals, and shall not be less than 1201 except in start-up and in final lots. Lot sizes shall be in multiples of pallet loads plus samples for destructive tests.

4.4.2 Inspection lot sample. - The inspection lot sample shall be selected and examined to the data package in accordance with MIL-STD-105, Inspection Level II, with an Acceptable Quality Level (AQL) of 1.0 defects per hundred units for MAJOR characteristics and 2.5 defects per hundred units for MINOR characteristics, except as specified in 4.4.3. One CRITICAL defect shall reject the lot.

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4.4.3 Sampling for performance tests. - After acceptance of the inspection lot sample, the performance lot sample shall be selected and tested in accordance with 4.5.1 and 4.5.2 under the conditions listed in Table II, in accordance with MIL-STD-105, inspection level S-3 for each test Condition with an AQL of 4.0 percent defective for MAJOR characteristics. One CRITICAL defect shall reject the lot.

TABLE II. PERFORMANCE TEST CONDITIONS

Test condition No.	SUS depth setting	Aircraft launch	
		Altitude ± 10% (ft)	Velocity ± 10% (knots, true air speed)
1	Shallow	300	300
2	Deep	300	300
3	Shallow	10,000	190
4	Deep	10,000	190

4.4.3.1 Performance lot sample inspection. - Prior to conducting the performance tests, 100 percent of the sample shall be visually examined to verify the following, with an AQL of 1.0 defects per hundred units for MAJOR characteristics and 2.5 defects per hundred units for MINOR characteristics (not annotated). One CRITICAL defect shall reject the lot.

(a) The container marking is legible and correct.

(b) The container is not deformed in a manner that would adversely affect sealing or removal of the contents.

(c) The lid gasket shall not be damaged or exhibit deterioration or permanent deformation.

(d) The presence of eight desiccant units, and one each humidity indicator, instruction sheet and ammunition data card.

(e) The styrofoam trays are anti-static treated (C3).

(f) There is no evidence of TNT exudate from the tail and the plastic closure plug is securely in place (C4). (Note: This applies to the SUS Mk 61 only.)

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(g) The arming wire assembly is properly assembled and not free to fall out by gravity (C5).

(h) The explosive section is not damaged (C6).

(i) The signal marking and color coding are legible and correct (C7).

(j) The depth setting wire is properly installed (M105).

(k) The four screws holding the A&F mechanism to the explosive section are tightened to flatten the lockwashers (M106).

(l) There is no evidence of corrosion of the metal parts.

(m) The tail shroud is not loose (M107).

4.5 Test and inspection methods.

4.5.1 Time from water entry to detonation. - The signal shall be launched from a suitable aircraft and the time from water entry to detonation measured to an accuracy of 1.0 percent of the mean specified times.

4.5.2 Detonation. - The detonation first bubble period shall be measured to an accuracy of 2.5 percent of the mean specified times.

4.5.3 Packaging. - Prior to shipment, the pallet loading shall be inspected to ensure the assemblage is complete, that markings for identity and shipping are in accordance with the applicable drawings and Section 5.

5. PACKAGING

5.1 Packing. - The signal shall be packed in ammunition boxes in accordance with the drawings and specifications listed on either DL 2127915 or DL 1332255.

5.2 Palletizing. - The packaged signals shall be palletized in accordance with the requirements specified on either WR 54/131 or WR 54/167.

5.3 Marking. - In addition to any special marking required by the contract or order, each exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - The signal is an expendable item intended for launching from aircraft or surface vessels to provide a powerful source of underwater acoustic energy. It may be used for echo ranging, position fixing, communication, bottom mapping, sound transmission studies, etc.

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6.2 Ordering data. - Procurement documents shall specify the following:

(a) Title, number, revision and date of this specification.

(b) Mark and Mod of signal and the applicable data list number required (see 1.2, 3.1 and 3.4). The packaging data list number and the palletizing WR number (see 5.1 and 5.2).

(c) Provisions for submission of first article samples (see 3.5 and 4.3).

(d) Provisions for submission of quality conformance samples for performance (destructive) testing and test activity address (see 4.4.3).

(e) The maximum time allowed for the Government tests for the first article and the quality conformance test samples.

(f) The material, quantity, schedule of delivery and providing activity for the Government furnished property (see 3.2).

(g) Identification of the cognizant activities (including codes) that represent the Government.

(h) Invoke the provisions of 7-104.79(a) of the Armed Services Procurement Regulations, Safety Precautions for Ammunition and Explosives, when not performed by a Government activity. Specify the specific sections to be complied with in DOD 4145.26M (see 6.3).

6.3 Safety precautions. - The safety precaution requirements of the "DOD Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M, are applicable. Note: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5, should be made applicable.

6.4 Specific safety precautions. - This is an explosive loaded item and therefore potentially very destructive. For specific safety precautions see Description, Operation and Handling Instructions - Signals, Underwater Sound (SUS), NAVAIR 11-1-107.

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6.5 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specifications.

(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications.

(c) Referenced Specification, Standards, and Requirements - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

(d) Drawings - Drawings or lists of drawings referenced herein shall have precedence over all subsidiary drawings. All referenced drawings shall apply to the extent specified thereon.

6.6 Changes from previous issue - Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 This specification is under the cognizance of AIR-53301G.

Custodian:
Navy (AS)

Preparing Activity:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-S-23311A(AS) Signal, Underwater Sound, Mark 61 Mod 0 and Mark 64 Mod 0

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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